Proposal Report

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Executive Summary

Educational assessment data is frequently reported as average scores, omitting crucial context and leading policymakers to attribute achievement gaps to demographic groups rather than systemic inequities. We propose an interactive dashboard using Programme for International Student Assessment (PISA) data that displays full score distributions alongside contextual variables like socioeconomic background, guiding users toward fairer interpretations and reducing the risk of biased policymaking.

Introduction

The Problem and Why It Matters

Relying solely on average scores to compare student groups obscures critical information: it hides within-group variation and ignores structural inequities such as disparities in school resources. Research shows this approach causes audiences to attribute performance gaps to intrinsic group traits rather than systemic factors (Holder and Xiong 2022), with direct consequences for how education departments allocate resources. Students most in need of support are the most likely to be mischaracterised and underserved.

Aligning with ETS's mission regarding fairness and equity in assessment (Educational Testing Service 2026), this project builds on the success of ETS's existing NAEP dashboard to demonstrate that better data presentation leads to better policy.

Background

PISA is the largest comparative education study globally. Administered by the OECD every three years, it assesses 15-year-old students across 81 countries (OECD 2023). Beyond cognitive scores in mathematics, reading, and science, PISA collects extensive contextual data on socioeconomic status and learning environments.

Specific Objectives

We will build an interactive, scaffolded dashboard that displays full score distributions with contextual filters, enabling nuanced comparisons over time and across comparable countries. The tool aims to replace deficit thinking with contextually grounded interpretations of student data — accessible to policymakers without statistical expertise. As a stretch goal, we will integrate an AI-assisted interface for natural language querying.

Final Data Product

The deliverable is an interactive Python application built with Streamlit, incorporating PISA data from 2015 to 2022.

Data & Exploratory Data Analysis (EDA)

Data Source, Size and Structure

Our primary data source is the PISA dataset. Our working sample, provided by ETS, covers Canada and the United States for the 2022 cycle — the foundation for our pipeline before expanding to all 81 countries across the 2015, 2018, and 2022 cycles.

The working dataset contains 27625 students and 1699 columns in a merged student-and-school-level file. Each row corresponds to one student, the **observational unit** for all analyses. After removing columns with non-trivial missingness and consulting with our partners, we narrowed the feature space to 132 columns across four categories, summarised in Table 1.

Table 1: Variable Categories and Key Columns

Category	Key Variables	Description
Identifiers and weights	`CNT`, `CNTSCHID`, `W_FSTUWT`, `W_FSTURWT1`–`W_FSTURWT80`	Student/school identifiers, primary sampling weight, 80 replicate weights for standard error estimation.
Cognitive outcomes	`PV1MATH`–`PV10MATH`, `PV1READ`–`PV10READ`, `PV1SCIE`–`PV10SCIE`	Ten plausible values per subject — statistically drawn estimates of latent ability.
Grouping variables	`ST004D01T`, `IMMIG`, `ESCS`	Gender, immigration background, and socioeconomic status index for distributional comparisons.
Contextual variables	`BELONG`, `MATHMOT`, `ST062Q01TA`	Student attitudes, behaviours, and school environment variables.

Statistical Complexity

Student proficiency is not reported as a single score. Each student receives 10 plausible values per subject derived through Item Response Theory scaling (OECD 2024, chap. 11). Correct estimation requires averaging across all ten plausible values, each weighted by `W_FSTUWT`. Canada's weighted mean math score is 496.9 points; the United States' is 464.9 points. Standard errors require jackknife resampling via 80 replicate weights, as students are sampled within schools (OECD 2024, chap. 10).

Exploratory Data Analysis

Finding 1: Quantifying What the Average Hides

Figure 1 shows the full math score distributions for Canada and the United States on a shared scale. While Canada's mean exceeds the USA's by 32.0 points, the distributional view reveals more: the gap at the 25th percentile (34.8 points) is larger than the mean gap, while the gap at the 75th percentile narrows to 31.0 points. This widening at the lower end is entirely invisible in a bar chart of averages, yet has direct policy implications (Malkus 2023).

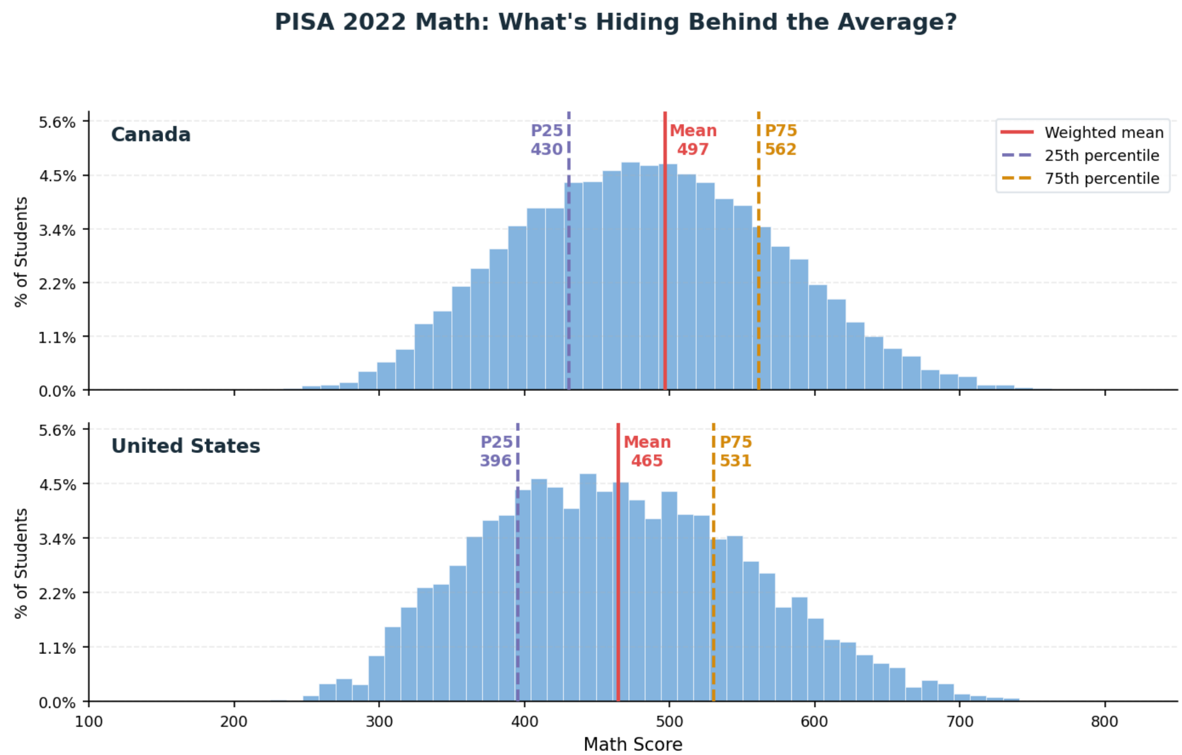

Figure 1: PISA 2022 math score distributions for Canada and the United States, with weighted mean, 25th and 75th percentile markers.

Finding 2: Gender Gap Across the Distribution

Figure 2 shows the gender gap (male minus female) at key percentiles across all three subjects. In mathematics, males score higher at every percentile, with the gap widening to approximately 25–30 points at the 90th percentile in both countries. The pattern reverses in reading, where females consistently outscore males across the full distribution.

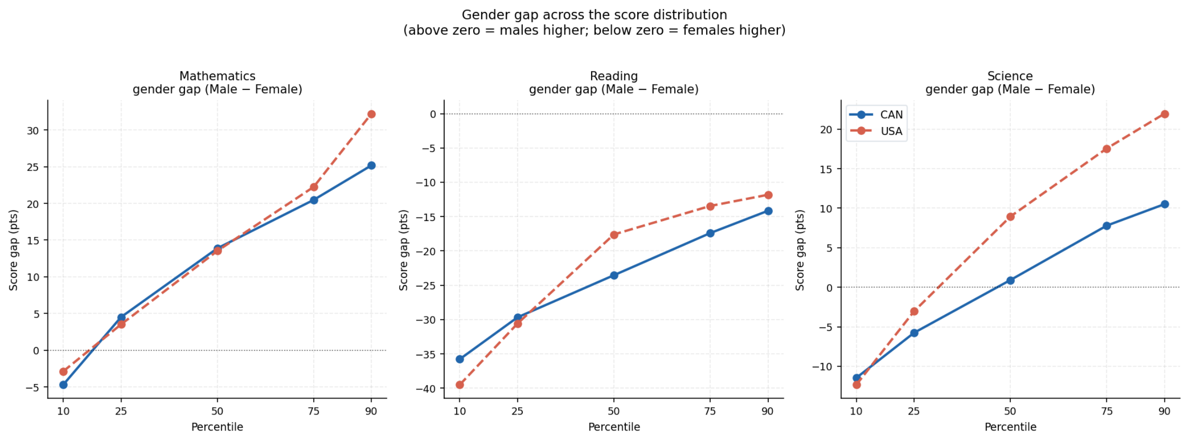

Figure 2: Gender gap (male minus female score) at the 10th, 25th, 50th, 75th, and 90th percentiles for mathematics, reading, and science — Canada and USA.

Finding 3: Socioeconomic Gap Across the Distribution

Figure 3 displays math score distributions by ESCS quartile. Distributions shift systematically rightward from Q1 (lowest) to Q4 (highest), consistent with the OECD finding that students in the top ESCS quartile score 93 points higher in mathematics than those in the bottom (OECD 2023, chap. 4) — the primary motivation for including ESCS as a dashboard filter.

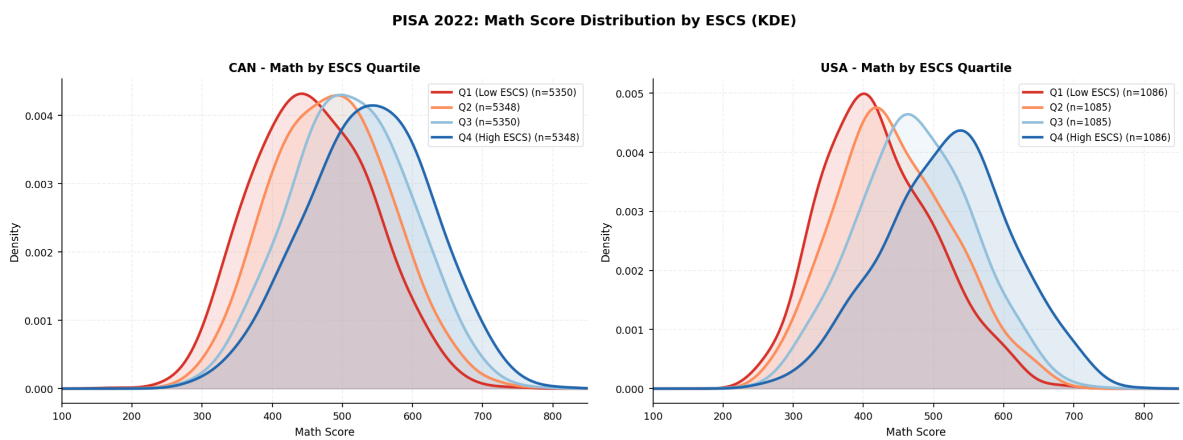

Figure 3: Math score distributions by ESCS quartile (KDE) for Canada and the United States.

Data Limitations and Challenges

Missingness is the primary data quality concern, summarised in Table 2. Columns from optional parent and wellbeing questionnaires are 100% missing in our sample and were dropped entirely. Comparability across cycles is a secondary challenge, as PISA occasionally revises questionnaire items (OECD 2024, chap. 20). Finally, the unequal sample sizes — Canada (23073 students) vs. the United States (4552 students) — require strict adherence to sampling weights.

Table 2: Missingness in Key Contextual Variables

	Variable	Missing Values	Missing (%)
0	Gender (ST004D01T)	47	0.2%
1	School type (SC001Q01TA)	1789	6.5%
2	Language at home (ST022Q01TA)	1816	6.6%
3	ESCS index	1887	6.8%
4	Immigration background (IMMIG)	3008	10.9%

Data Science Approach

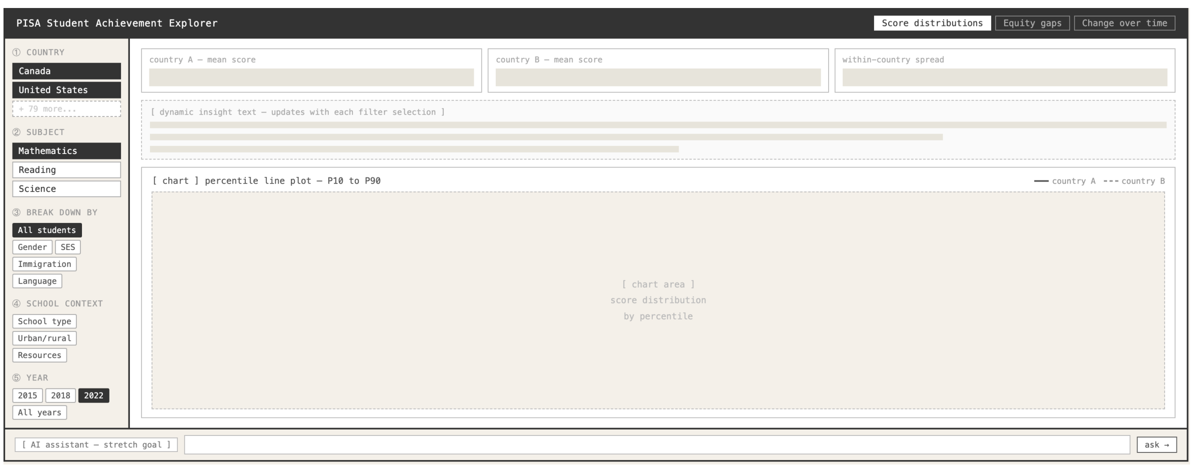

Figure 4: System blueprint and filtering architecture for the interactive dashboard.

Baseline

Our baseline is the full, weighted OECD score distribution per country, computed using sampling weights averaged across all ten plausible values. This provides a statistically valid benchmark for evaluating all subsequent subgroup comparisons.

Interactive Filtering and Visualisation

We will implement a two-tiered filtering system: primary filters (country, year, subject, gender) and contextual filters (ESCS, school resources, immigration background, sense of belonging). These will dynamically update Kernel Density Estimation (KDE) distributions and percentile line plots, overlaying contrasting groups rather than presenting isolated metrics.

Visualising full distributions prevents audiences from attributing performance gaps to intrinsic deficits — directly addressing Simpson's Paradox, where whole-population summaries mislead. Incorporating contextual variables enables fair comparisons between students with similar backgrounds, allowing policymakers to identify targetable structural factors.

Anticipated Challenges

Computing weighted distributions dynamically across millions of rows risks dashboard latency; we will mitigate this by pre-computing aggregated percentile bounds where possible. Contextual variables are also sparsely populated outside North America, requiring robust fallback handling to prevent visual errors.

Evaluation & Success Criteria

Success will be evaluated across four dimensions:

1. **Statistical accuracy:** All figures must use weighted means and percentiles averaged across all 10 plausible values, validated against published OECD statistics.
2. **Equity story visibility:** Breakdowns by socioeconomic status, gender, and immigration status must reveal distinct, interpretable distributional patterns.
3. **Usability by non-experts:** An educator or policymaker must reach meaningful insights without reading technical documentation, evaluated through walkthroughs with ETS partners.
4. **Partner sign-off:** ETS must confirm the design avoids deficit framing by prioritising contextualised score distributions.

Timeline

Work is organised into four two-week phases:

- **Phase 1 (Weeks 1–2):** Finalise variable selection with ETS and build the core statistical engine, validated against OECD figures before UI development begins. Fallback: if equity variables lack global coverage, we will use a minimum viable layer of ESCS and gender.

- **Phase 2 (Weeks 3–4):** Build the first working dashboard for Canada and the United States, including page structure, filter interfaces, and core percentile charts.
- **Phase 3 (Weeks 5–6):** Integrate the equity layer — SES, gender, and immigration breakdowns alongside school context overlays and dynamic insight text.
- **Phase 4 (Weeks 7–8):** Scale to all 81 PISA countries, finalise interactivity, incorporate partner feedback, and attempt the AI chatbot stretch goal.

References

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